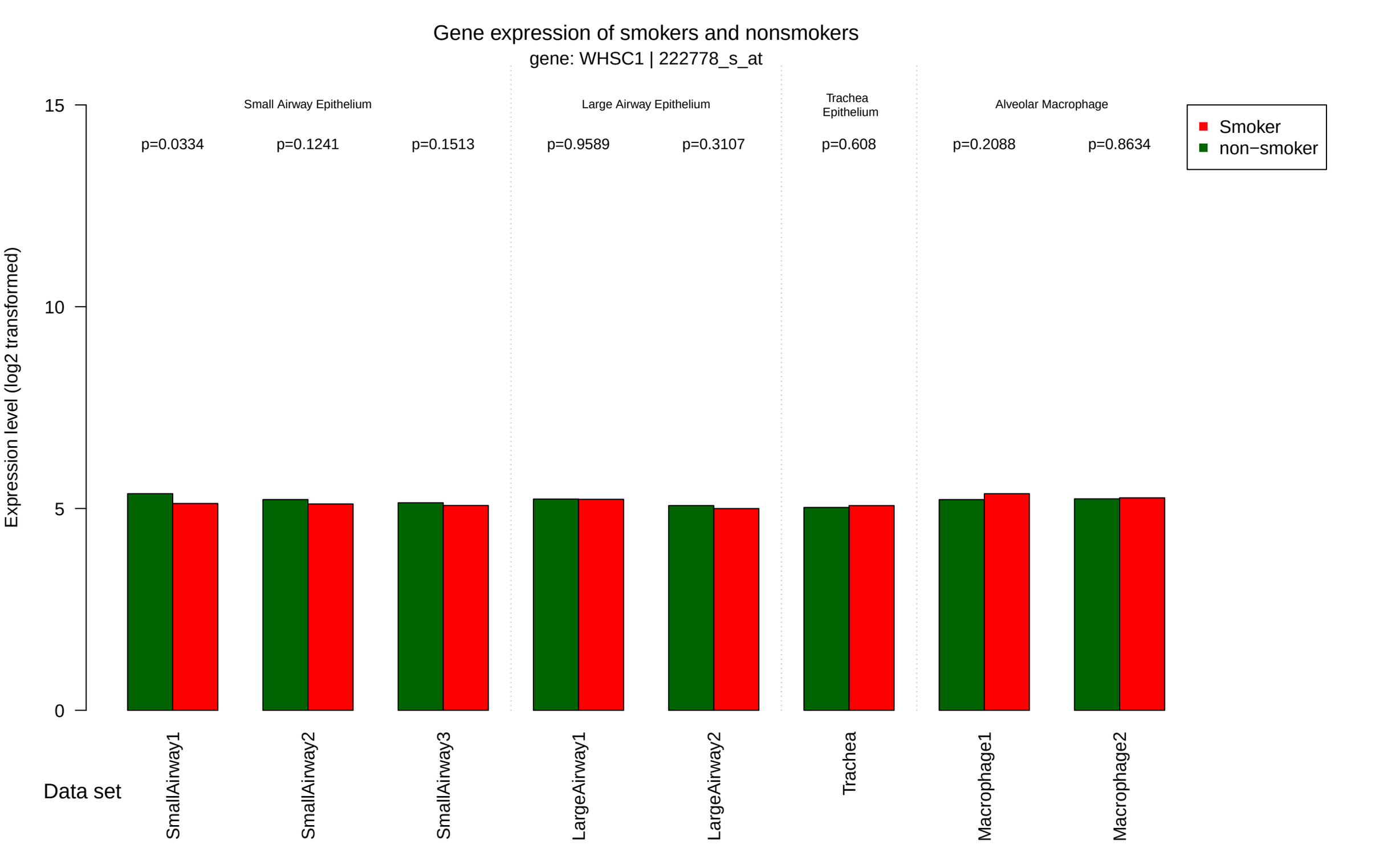
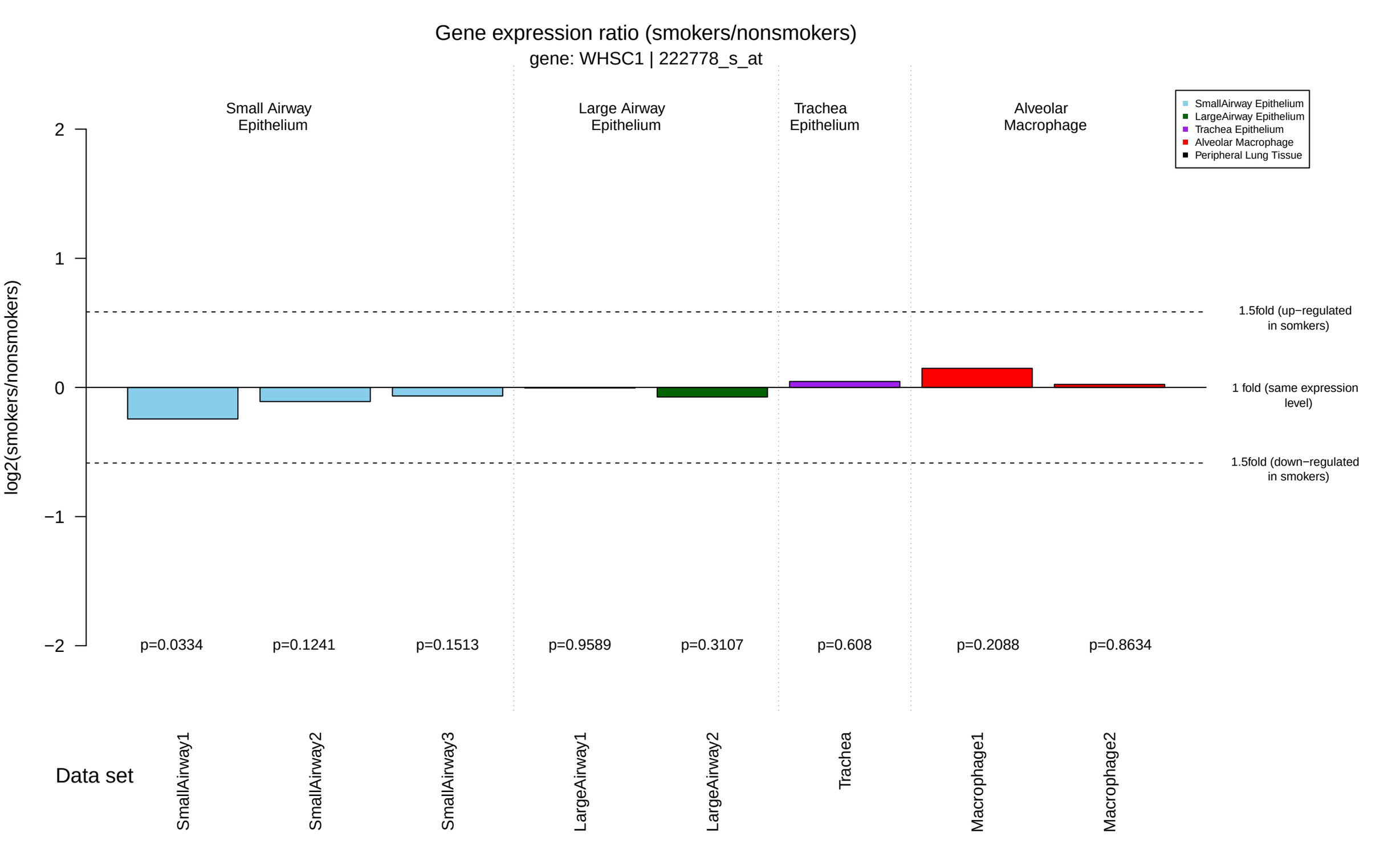
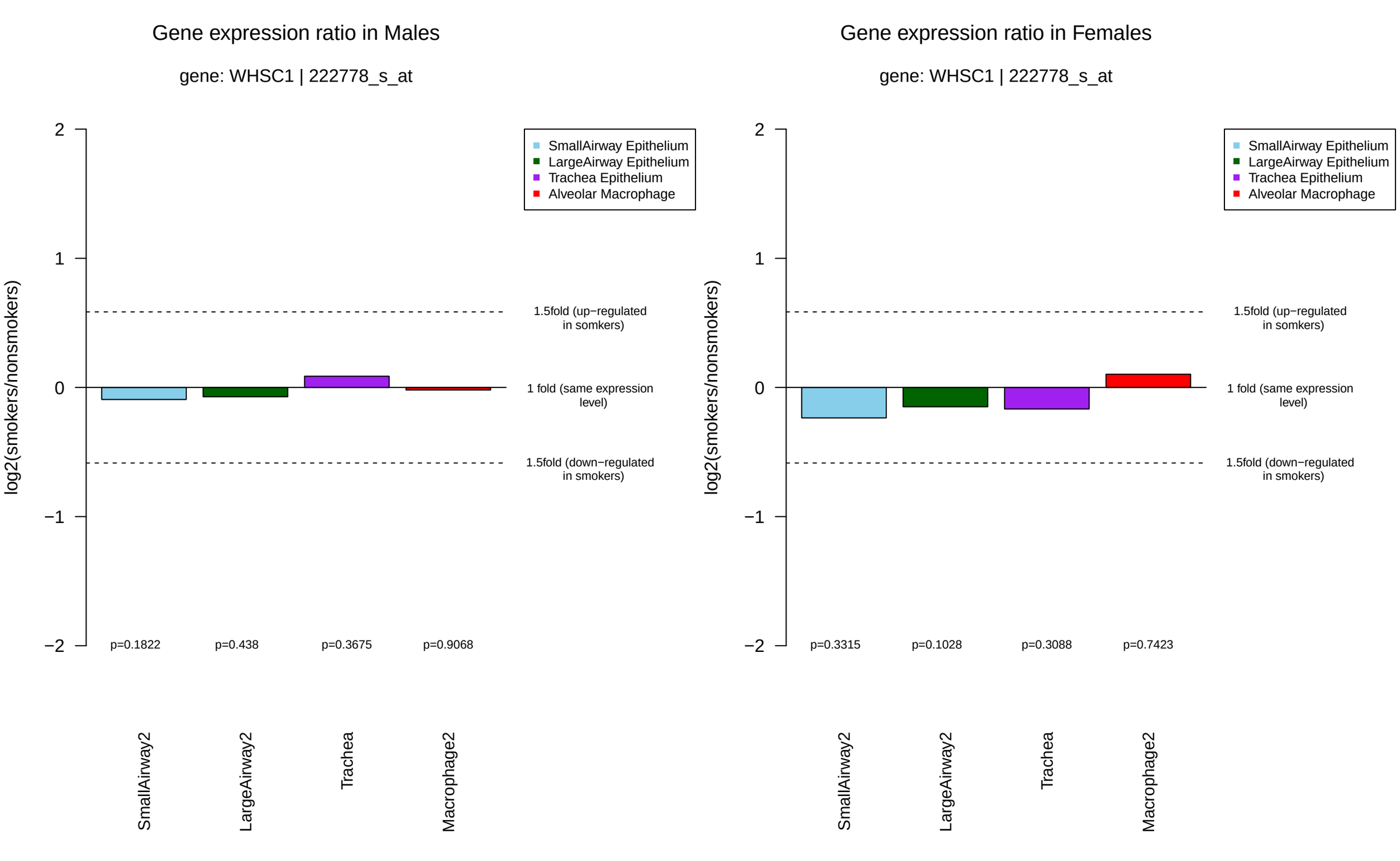
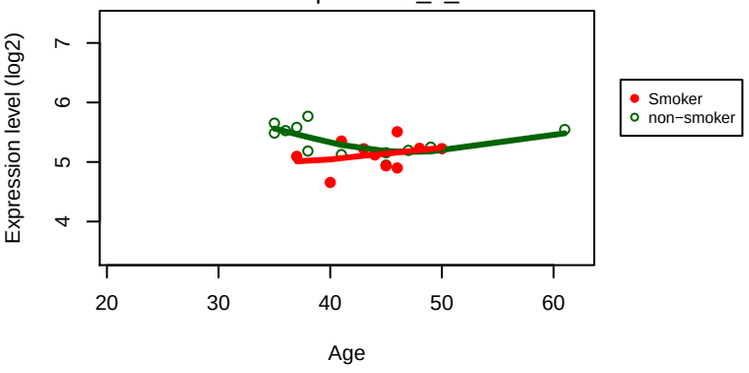
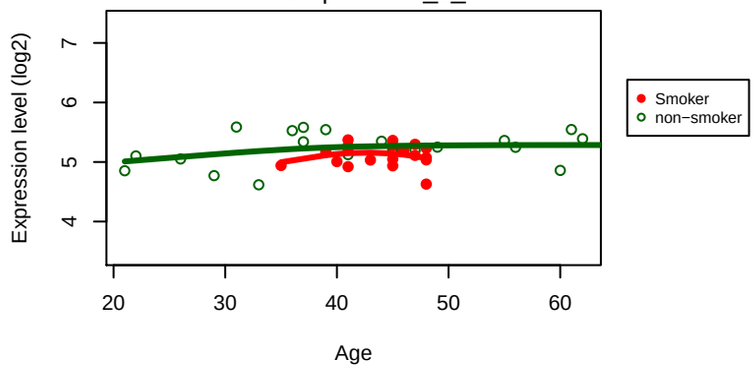
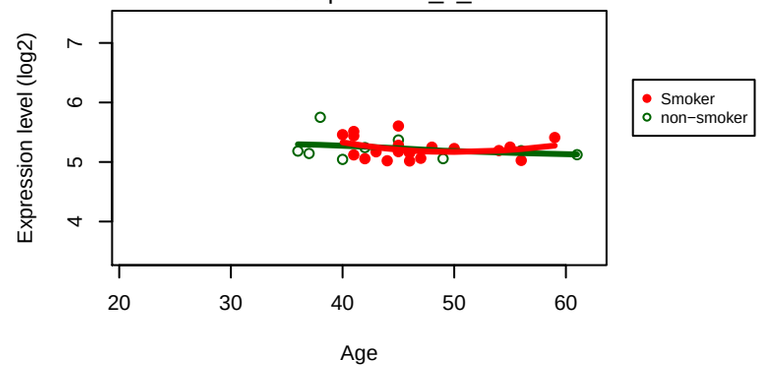

Gene expression of smokers and nonsmokers
gene: WHSC1 | 222778_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0334
p=0.1241
p=0.1513
p=0.9589
p=0.3107
p=0.608
p=0.2088
p=0.8634
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

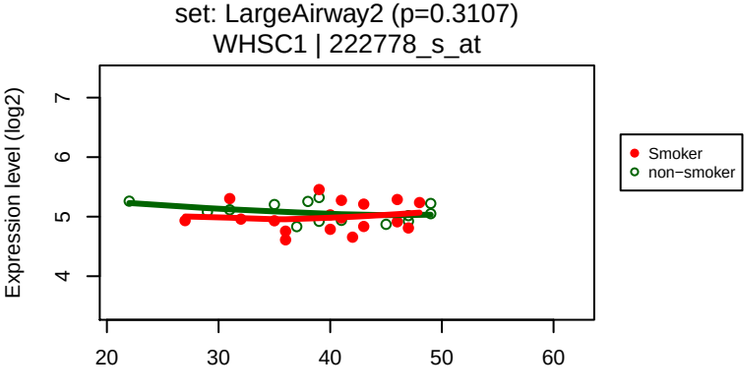

Gene expression ratio (smokers/nonsmokers)
gene: WHSC1 | 222778_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0334
p=0.1241
p=0.1513
p=0.9589
p=0.3107
p=0.608
p=0.2088
p=0.8634
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

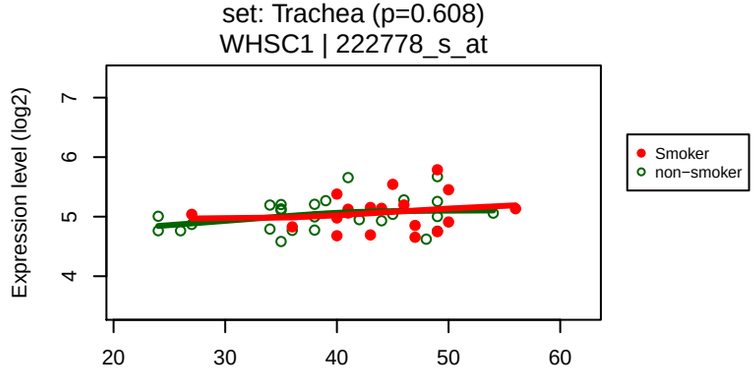

Gene expression ratio in Males
gene: WHSC1 | 222778_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1822
p=0.438
p=0.3675
p=0.9068
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WHSC1 | 222778_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3315
p=0.1028
p=0.3088
p=0.7423
SmallAirway2
LargeAirway2
Trachea
Macrophage2

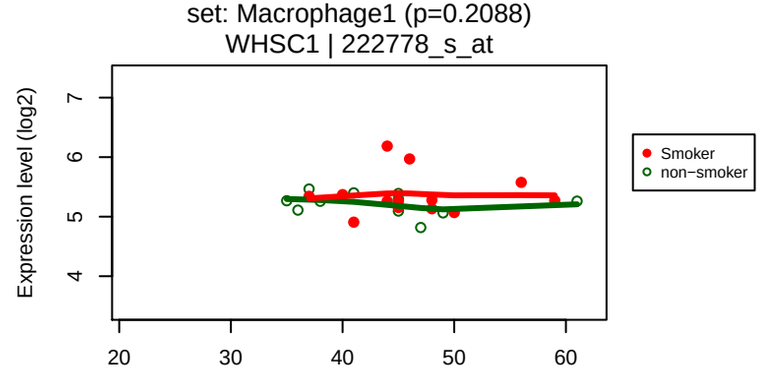

set: SmallAirway1 (p=0.0334)
WHSC1 | 222778_s_at
Expression level (log2)
Age
Smoker
non−smoker

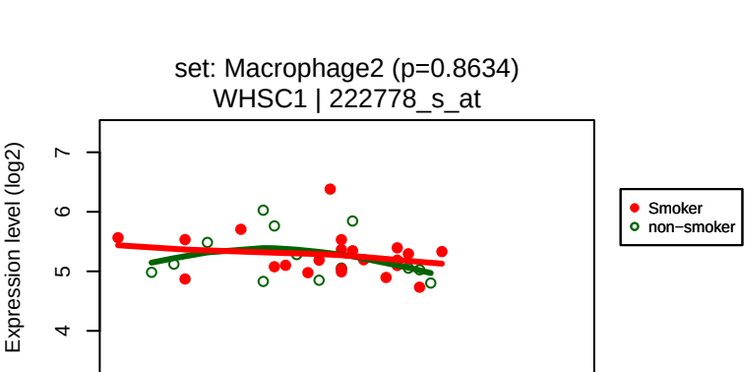

set: SmallAirway2 (p=0.1241)
WHSC1 | 222778_s_at
Expression level (log2)
Age
Smoker
non−smoker

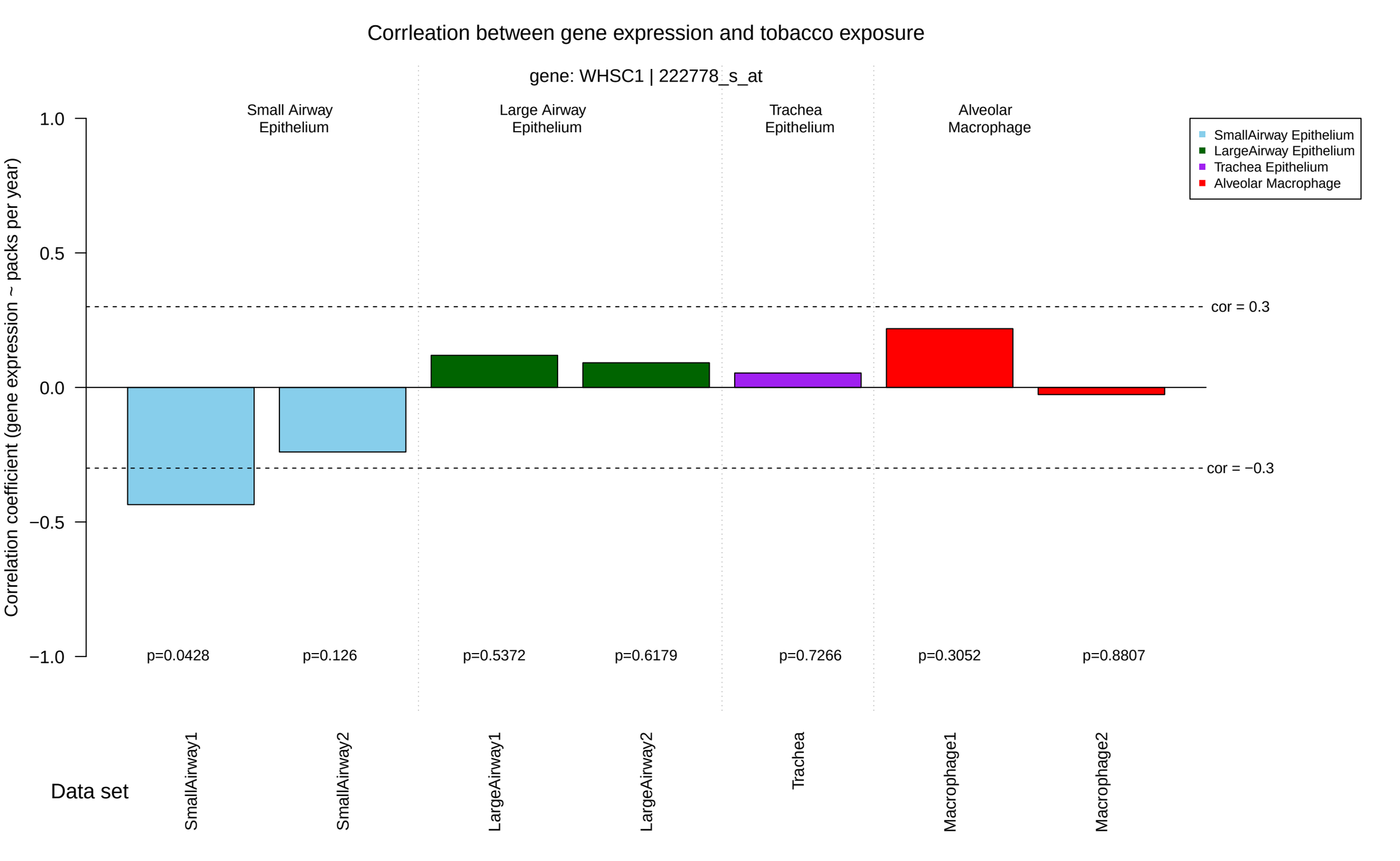

set: LargeAirway1 (p=0.9589)
WHSC1 | 222778_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3107)
WHSC1 | 222778_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.608)
WHSC1 | 222778_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2088)
WHSC1 | 222778_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8634)
WHSC1 | 222778_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WHSC1 | 222778_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0428
p=0.126
p=0.5372
p=0.6179
p=0.7266
p=0.3052
p=0.8807
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2